Customer Churn Prediction Model

Apollo Diagnostic Focus

Identifying and Winning Back At-Risk Patients

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Executive Summary

The Challenge

High-value chronic patients stopping routine tests, leading to lost recurring revenue.

The Solution

Churn prediction model with behavioral analytics and a win-back strategy.

The Impact

Personalized Health Reminder campaigns to retain customers.

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Professional Strengths & Mindset Framework

The 'Mindset' Framework

Business Analyst (9.5/10)

Focus on ROI and saving a real family business rather than just coding

Data Analyst (9/10)

Identifying the need for Qualitative Data (Feedback Forms) to supplement missing Quantitative Data

ML Engineer (7/10)

Using behavioral features to predict the Target Variable (Churn vs. Retained)

Fresher Value (10/10)

Solving a unique, real-world local problem instead of using generic datasets

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Data Collection Strategy

Internal Data Sources

- Patient ID
- Last Visit Date
- Test Type
- Bill Amount
- Home Collection vs. Walk-in

External Data Sources (Feedback)

4-question survey focused on:

- Price Sensitivity
- Service Speed
- Convenience

Days_Since_Last_Test — the primary indicator of whether a patient has already 'churned.'

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Technical Roadmap for Wipro-Level Standards

Step 1: Define Churn

No visit for >180 days for chronic patients

Step 2: Algorithm Selection

Compare Logistic Regression (to explain why they leave) with Random Forest/XGBoost (for high accuracy)

Step 3: Explainability

Use SHAP values to tell the business manager exactly why a specific patient was flagged as high-risk

Step 4: Deployment

Create a simple Streamlit dashboard to display "At-Risk" patients for the center's staff to call

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Interview Impact Pitch

"I observed my family's diagnostic business losing repeat customers. To solve this, I designed a feedback-led churn model. I didn't just predict who would leave; I analyzed the root causes—like wait times and pricing—to suggest a personalized 'Health Reminder' win-back strategy."

Real-world problem solving

Not just academic, but solving an actual family business challenge

Data-driven insights

Combining quantitative and qualitative data for actionable recommendations

Business impact

Personalized retention strategy to recover lost revenue

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Conclusion: Driving Retention, Ensuring Growth

This project successfully addresses the critical business challenge of patient churn in diagnostic services. By leveraging advanced machine learning and behavioral insights, we've developed a robust solution that not only predicts patient attrition but also empowers proactive engagement, securing recurring revenue and fostering patient loyalty.

Business Acumen

Focused on ROI, real-world problem-solving, and integrating qualitative feedback for actionable insights.

Technical Excellence

Demonstrated through rigorous algorithm selection, explainable AI (SHAP), and practical deployment via Streamlit.

Sustainable Impact

Delivering a personalized "Health Reminder" strategy that directly contributes to patient retention and business sustainability.

This model establishes a scalable framework for personalized patient care, offering a blueprint for expansion across multiple diagnostic centers and potential application in broader healthcare predictive analytics.

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